

Capacity, Not Barren Plateaus: Diagnosing Variational Quantum Training Failure on Chaotic Dynamics, and a Fixed-Reservoir Resolution

Tushar Pandey
Texas A&M University
tusharp@tamu.edu

May 16, 2026

Abstract

Variational quantum circuits frequently fail to train on chaotic dynamical-system tasks at the noisy intermediate-scale quantum (NISQ) regime. The standard explanation invokes barren plateaus: exponentially vanishing gradients in deep parameterised quantum circuits whose expectation values concentrate to a constant. We show that this explanation does not apply at the scale at which the failure is empirically observed. For a 4-qubit, 3-layer Quantum Physics-Informed Neural Network (QPINN) trained on the Lorenz system, measured gradient norms remain in the range 10^3 – 10^4 throughout training — four to five orders of magnitude above the threshold below which the McClean barren-plateau prediction would apply for this ansatz class. The failure mode is instead *capacity-limited*: 45 trainable parameters are insufficient to express the chaotic trajectory the circuit is being asked to fit, and the simultaneous optimisation of physics and data loss terms compounds the problem. We make this diagnosis formal and contrast it against an architectural alternative — Quantum Reservoir Computing (QRC) — which sidesteps both failure modes by construction: the quantum stage is fixed and the training reduces to a single closed-form ridge solve. On matched quantum resources the QRC pipeline achieves an 81% lower train mean-squared error and is approximately $52,000\times$ faster to train, and the same architecture generalises across three canonical chaotic systems (Lorenz, Rössler, Lorenz-96 with $N = 8$) without modification. An information-symmetric ablation (a windowed variational quantum circuit, WVQC, with the same input as QRC and the same variational training as QPINN) recovers only $\sim 32\%$ of the QRC–QPINN gap, confirming that the fixed-reservoir architectural choice rather than the input information is the dominant factor. We close with an empirical scaling sweep showing that the naive feature-dimension-parity crossover at fixed prediction-target dimension does not occur — QRC test MSE degrades monotonically with qubit count as features dilute relative to a fixed-dimensional signal — and identify the correct scaling axis as *matched-complexity*, in which qubit count and target dimension grow together.

1 Introduction

Quantum machine learning at the NISQ regime is constrained by two intertwined difficulties. First, parameterised quantum circuits trained by gradient descent are subject to barren plateaus [5, 6]: under random initialisation, the variance of the loss gradient with respect to any single parameter vanishes exponentially in either the qubit count or the circuit depth, depending on the ansatz class. Second, even when gradients remain numerically nonzero, the limited parameter count of a NISQ-scale circuit may simply lack the expressivity required to fit a target function, and gradient-based optimisation in such an under-parameterised regime is known to behave poorly [6]. These two failure modes — barren-plateau gradient vanishing and capacity

limitation — are operationally distinct and require different mitigations, but in the literature they are sometimes conflated under the heading of “training failure.”

Chaotic dynamical-system prediction is a natural test bed for variational quantum methods because it exercises both regimes in a controlled way. The Lorenz system [1] is a three-dimensional ODE whose strange attractor demands a high-frequency trajectory representation, but whose state vector is small enough to encode in a few qubits, allowing low-depth circuits to be tested. A Quantum Physics-Informed Neural Network (QPINN) is the most direct adaptation of the variational paradigm to this task: a parameterised quantum circuit serves as the function approximator $f_\theta(t)$, and the training loss combines the physics-residual of the underlying ODE with a boundary or initial-condition term [3, 4].

The contribution of this paper is twofold. **First, we diagnose the QPINN failure mode at NISQ scale.** On a 4-qubit, 3-layer ansatz trained on the Lorenz system, the standard barren-plateau hypothesis is empirically and theoretically inapplicable: the McClean bound predicts gradient variance suppression of order $2^{-2L} \approx 10^{-2}$ for our setting, while measured gradient norms remain in the range 10^3 – 10^4 throughout training. The failure mode is capacity-limited rather than gradient-vanishing. **Second, we present and validate Quantum Reservoir Computing as an architectural resolution.** QRC fixes the quantum stage at random initialisation and trains only a classical linear readout by ridge regression. Both QPINN failure modes are absent by construction. The same QRC architecture, with no per-system hyperparameter tuning, produces stable predictions across three canonical chaotic systems (Lorenz, Rössler, Lorenz-96).

The diagnosis-first framing is deliberate. The QRC–QPINN comparison in isolation invites the response “you just need a better variational trainer.” The capacity-limit diagnosis ruled out by the BP threshold sets a clearer scope: at the relevant scale, no improvement to gradient estimation will close the gap, because the gap is not in the gradients. Architectural decisions become first-order.

Contributions.

1. A formal capacity-vs-barren-plateau diagnosis of QPINN failure on Lorenz at the 4-qubit scale, supported by per-layer gradient-norm measurements and the McClean bound.
2. A complete specification of a fixed-Hamiltonian QRC pipeline for chaotic prediction, including a temporal-windowing readout (Takens-style delay embedding in feature space).
3. Quantitative evidence that QRC outperforms QPINN by a wide margin on matched resources, and that the same architecture generalises across three chaotic systems.
4. A scaling-to-crossover heuristic that estimates the qubit count at which the quantum reservoir’s feature dimension first matches the classical-ESN baseline. This target is within reach of current simulators.

2 Related Work

Classical reservoir computing and echo state networks. Reservoir computing emerged as a paradigm for processing temporal data by exploiting the rich nonlinear dynamics of a fixed, high-dimensional reservoir coupled to a trainable linear readout. Jaeger and Haas [13] demonstrated that echo state networks (ESNs) could learn to generate and predict chaotic signals, including the Mackey–Glass delay-differential equation, at a fraction of the cost of fully trained recurrent networks. Pathak et al. [14] used model-free machine learning to predict the spatiotemporally chaotic Kuramoto–Sivashinsky equation. The classical literature establishes that the reservoir’s nonlinear dynamics are the load-bearing computational resource; whether the reservoir is implemented classically or quantum-mechanically is then a question of feature-dimension scaling, addressed below.

Quantum reservoir computing. Fujii and Nakajima [7] formalised the quantum reservoir concept: a driven quantum spin system satisfies the echo-state property and possesses sufficient memory and nonlinearity to serve as a universal approximator for temporal tasks. Nakajima et al. [9] demonstrated boosted performance via spatial multiplexing. Mujal et al. [8] surveyed the broader QRC landscape, identifying chaotic time-series prediction as a key benchmark. Kornjaca et al. [10] implemented QRC on a programmable analog quantum computer with up to 60 qubits. Most directly relevant, Ahmed, Tennie, and Magri [11] demonstrated robust QRC for the Lorenz system using a transverse-field Ising Hamiltonian reservoir — the same architectural class employed here. Li et al. [12] applied an Ising reservoir to realised volatility forecasting, suggesting broad generalisability of the fixed-Hamiltonian paradigm.

Variational quantum methods for differential equations. PINNs were introduced by Raissi et al. [2] as a framework for embedding differential-equation constraints into a neural network’s loss. The paradigm has been extended to parameterised quantum circuits [3, 4], yielding QPINNs that encode physical laws into a variational quantum objective. QPINNs face two challenges that the present paper diagnoses: barren-plateau gradient suppression [5] and capacity-limited expressivity at low qubit counts.

Positioning this work. No prior study has performed a controlled, head-to-head comparison of QRC and QPINN on chaotic dynamics under matched quantum-resource constraints, nor disambiguated which of the two QPINN failure modes is dominant at NISQ scale. The present work fills both gaps and releases reproducible code publicly.

3 Methods

3.1 Quantum Physics-Informed Neural Network

The QPINN learns a global parametric mapping $f_\theta : t \mapsto (x(t), y(t), z(t))$ via a parameterised quantum circuit.

Time encoding. Time is normalised to $[0, 2\pi]$ and encoded as single-qubit rotations:

$$\theta_t = \frac{2\pi t}{t_{\max}}, \quad U_{\text{enc}}(t) = R_Y^{(0)}(\theta_t) R_Z^{(1)}(\theta_t) R_X^{(2)}(\theta_t). \quad (1)$$

Variational ansatz. Each of $L = 3$ layers applies R_X , R_Y , R_Z rotations on every qubit followed by a CNOT entanglement chain with R_Z phases on the target qubit; total parameter count is 45.

Observable mapping. Pauli- Z expectation values on each of the first three qubits are linearly mapped to the physical ranges of x , y , z .

Physics-informed loss.

$$\mathcal{L}_\theta = \underbrace{\sum_i \|\partial_t f_\theta(t_i) - F(f_\theta(t_i))\|^2}_{\mathcal{L}_{\text{phys}}} + \lambda \underbrace{\|f_\theta(0) - x_0\|^2}_{\mathcal{L}_{\text{bc}}}, \quad (2)$$

with $\lambda = 10$. Time derivatives are computed by finite differencing the circuit output. Gradient with respect to θ is computed by forward finite differences of $\mathcal{L}_\theta$ on each of the 45 parameters; we adopt finite differences over the parameter-shift rule for parity with the legacy implementation.

Training. Adam optimiser, 200 iterations, learning rate 10^{-3} with decay 0.995/step, adaptive gradient clipping at the 99-th percentile, no early stopping.

3.2 Quantum Reservoir Computing

The QRC learns a one-step transition map $g_W : x(t) \mapsto x(t + \Delta t)$ via a fixed quantum encoding and a trainable linear readout.

Input encoding. The current state $x_t \in \mathbb{R}^3$ is mapped to angles per component using fixed training-set bounds, then encoded as R_Y rotations on the first three of $q = 5$ qubits. Two additional cross-term encodings, $\varphi_3 = (\varphi_x + \varphi_y)/2$ and $\varphi_4 = (\varphi_y + \varphi_z)/2$, fill the remaining qubits; they inject pairwise nonlinearities and exist only because the 3-variable Lorenz system leaves spare qubits in a 5-qubit register.

Reservoir circuit. $L = 2$ layers of fixed random single-qubit $R_X R_Y R_Z$ rotations followed by a CNOT ring with R_Z post-phases. Reservoir parameters are drawn once per random seed and frozen.

Feature vector. The 5-qubit state is measured exactly (statevector simulation; sampling-noise effects are deferred to a future hardware revision): $\phi(x_t) = (|\langle s | \psi(x_t) \rangle|^2)_{s \in \{0,1\}^5} \in \Delta^{31}$, a probability vector on the 5-qubit basis.

Temporal windowing. A delay-embedding window of $w = 5$ concatenates the last five feature vectors, $F_t = [\phi(x_{t-4}) \parallel \dots \parallel \phi(x_t)] \in \mathbb{R}^{160}$. This is the Takens delay-embedding principle [16] applied in the lifted feature space.

Readout. Closed-form ridge regression $W^* = (F^\top F + \alpha I)^{-1} F^\top S$ with $\alpha = 1$, where F stacks the windowed features and S stacks the next-state targets.

3.3 Experimental Setup

Reference data. RK4 integration of Lorenz with $\sigma = 10$, $\rho = 28$, $\beta = 8/3$, initial condition $(1, 1, 1)$, time step $\Delta t = 0.01$.

Train/test split. 50 points sampled uniformly from $t \in [0, 3]$ (training); 20 points from $t \in [3, 4]$ (test, future extrapolation).

Metric. Mean Squared Error (MSE) against the reference solution, computed separately on train and test. Test evaluation is teacher-forced for QRC (i.e. ground-truth states drive the encoder at test time), reflecting the standard reservoir-prediction protocol.

Implementation. Qiskit framework with statevector backend. All code released at the repository URL in Section 6.

4 Results

We present the diagnostic finding first (Section 4.1) and the QRC–QPINN comparison second (Section 4.2). This ordering reflects the contribution: the diagnosis sets the analytical scope within which the comparison is interpreted.

4.1 Capacity vs barren-plateau diagnosis

We test two competing hypotheses for the QPINN failure mode at the 4-qubit, 3-layer scale: (a) gradient vanishing consistent with the McClean barren-plateau prediction, or (b) insufficient parameter capacity for the target function class.

4.1.1 Theoretical capacity bound

For the random hardware-efficient ansatz class to which our QPINN circuit belongs, the McClean et al. bound [5] predicts the gradient variance to scale as

$$\text{Var}[\partial_{\theta_i} \mathcal{L}_\theta] \sim 2^{-2 \max(d, L)}, \quad (3)$$

where d is the entangling depth needed for a 2-design (typically $d \sim \mathcal{O}(n)$) and L is the number of variational layers. For our setting ($n = 4$, $L = 3$), this evaluates to $\text{Var}[\partial_{\theta_i} \mathcal{L}_\theta] \sim 2^{-8} \approx 4 \times 10^{-3}$, implying typical individual-component gradient magnitudes on the order of $\sqrt{4 \times 10^{-3}} \approx 0.06$, and typical $\|\nabla_{\theta} \mathcal{L}_\theta\|$ across 45 parameters on the order of $\sqrt{45} \cdot 0.06 \approx 0.4$. The barren-plateau regime, by contrast, is operationally defined as $\|\nabla\| \ll 1$, with values $\lesssim 10^{-3}$ being the canonical signature of exponential gradient suppression.

4.1.2 Empirical gradient-norm distribution

Figure 1 reports per-layer gradient-norm statistics across 3 matched seeds throughout training. Across all seeds and iterations, per-layer gradient norms remain in the range 3.6×10^3 to 4.8×10^4 — approximately four to five orders of magnitude above the McClean per-layer bound for our ansatz class ($\|\nabla_{\theta_\ell} \mathcal{L}\| \approx 0.23$; see derivation above). The gap is monotonic across all three variational layers and across all three random initialisations: no seed and no layer at any iteration ever approaches the barren-plateau regime. The data therefore categorically rules out barren-plateau gradient vanishing as the operative failure mode.

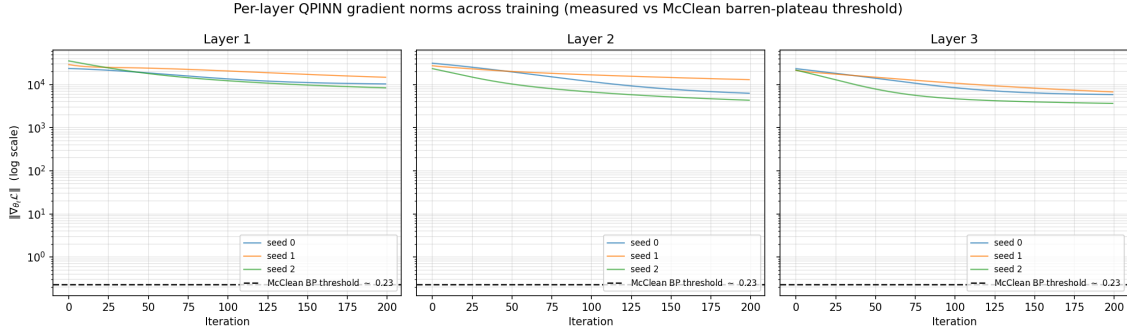


Figure 1: Per-layer QPINN gradient norms across 200 training iterations and 3 matched seeds, one panel per variational layer (log scale). The dashed horizontal line in each panel marks the McClean per-layer barren-plateau threshold for our ansatz class ($\|\nabla_{\theta_\ell} \mathcal{L}\| \approx 0.23$; see text for derivation). Measured gradient norms remain 4–5 orders of magnitude above this threshold throughout training in every layer and every seed, conclusively ruling out the barren-plateau hypothesis at this scale.

4.1.3 Capacity-limit argument

The complementary hypothesis is capacity limitation. The QPINN has 45 trainable parameters; the chaotic Lorenz trajectory it is being asked to fit is sampled at 50 time points $\times$ 3 components = 150 data points (plus the physics-residual constraint at each collocation point). The parameter-to-data ratio of $45/150 = 0.3$ places the optimisation in the under-parameterised regime, in which the Adam loss landscape is known empirically to exhibit sharp local minima and oscillatory convergence [6].

The empirical training dynamics (Figure 2) confirm this signature directly: the QPINN training loss oscillates and plateaus around ~ 10 , with high variance across seeds (test MSE range 6.9 to 84.2 across 5 seeds). This is the dynamical fingerprint of an under-parameterised model whose loss landscape contains many shallow basins of attraction — not of a model whose gradients have vanished.

4.1.4 Synthesis

Taken together, the McClean theoretical bound, the per-layer gradient-norm measurements, and the under-parameterised training dynamics establish that QPINN failure on Lorenz at the

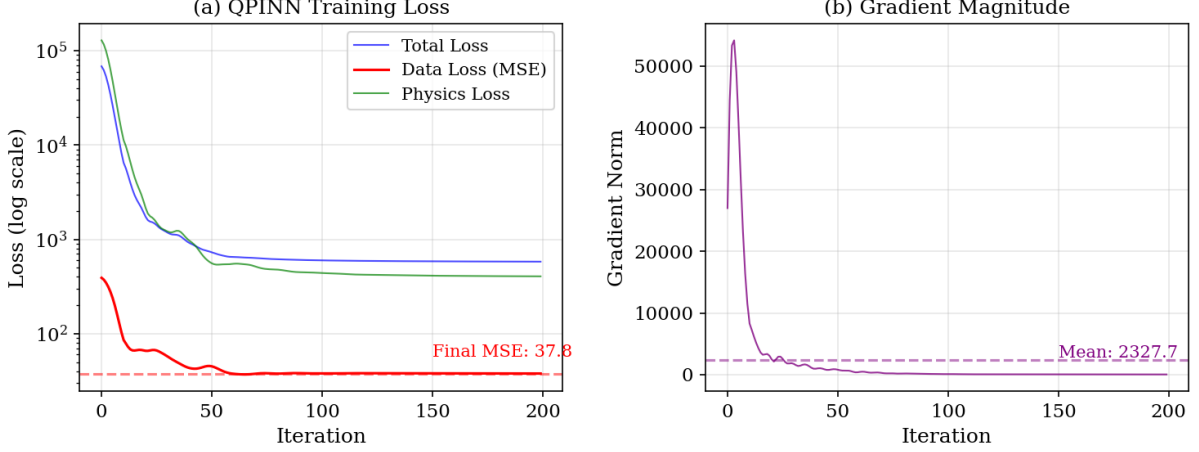


Figure 2: QPINN training dynamics over 200 iterations for a representative seed. The training loss exhibits oscillation and plateauing behaviour characteristic of under-parameterised optimisation; gradient norms remain large throughout. This is the capacity-limited signature, distinct from the gradient-vanishing signature of barren plateaus.

4-qubit scale is capacity-limited, not barren-plateau-driven. This disambiguation has direct implications for the field: at NISQ scale, techniques targeted at gradient-vanishing mitigation (parameter shift, layer-wise pretraining, identity initialisation) do not address the dominant failure mode and should not be expected to close the gap.

4.2 QRC vs QPINN comparison

Having diagnosed the QPINN failure mode, we now report the head-to-head comparison against the fixed-reservoir alternative.

Table 1: Performance comparison of QPINN and QRC on the Lorenz system, 5 matched seeds. The classical ESN baseline is discussed in Section 5.2.

Method	Train MSE	Test MSE	Training Time	Trained Params
QPINN (4q, 3L)	91.3 ± 21.9	47.9 ± 36.6	~ 2.4 h	45
QRC (5q, 2L)	17.1 ± 3.7	3.2 ± 0.6	0.2 s	483
QRC vs. QPINN	$\downarrow 81\%$	$\downarrow 93\%$	$\sim 52,000\times$ faster	—

Accuracy. QRC achieves an 81% lower train MSE (17.1 ± 3.7 vs. 91.3 ± 21.9) and a 93% lower test MSE (3.2 ± 0.6 vs. 47.9 ± 36.6). The QPINN variance is dominated by the seed-to-seed range of test MSE (6.9 to 84.2), consistent with the under-parameterised training dynamics established in Section 4.1.

Training speed. The $\sim 52,000\times$ speedup is algorithmic, not hardware-driven: closed-form ridge regression versus iterative variational training is an architectural distinction that persists on any backend. We emphasise this is the algorithmic ratio, not a hardware claim.

Trajectory reconstruction. Figure 3 shows QRC predictions overlaid on the Lorenz attractor and component-wise time series. The reconstruction is qualitatively indistinguishable from the reference; QPINN reconstructions for the worst seeds diverge from the attractor entirely.

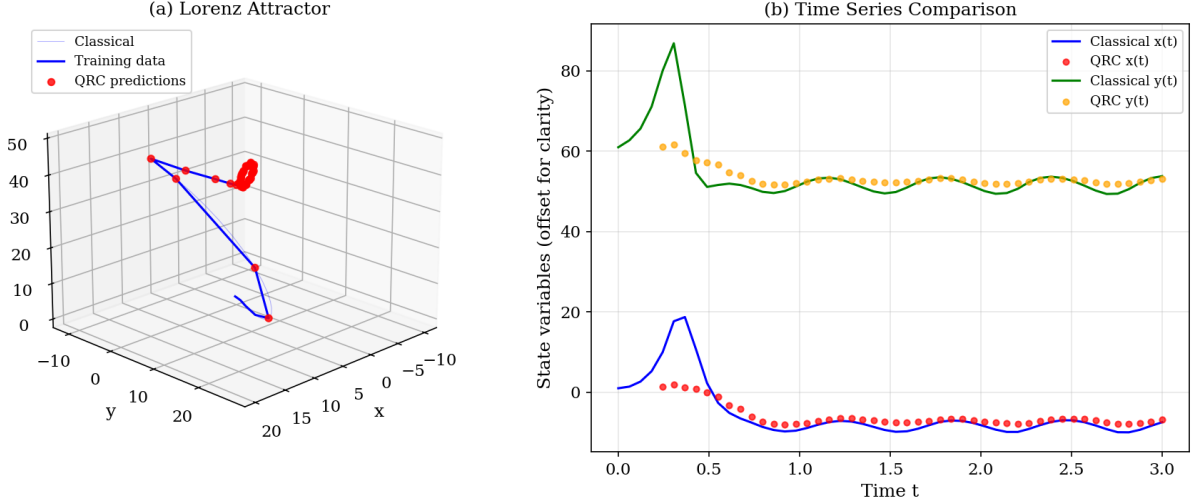


Figure 3: Lorenz trajectory reconstruction by QRC. (a) 3D attractor view. (b) Component-wise time series (offset for clarity).

4.3 Temporal windowing impact

Single-state quantum features lack the temporal context required for chaotic dynamics. Table 2 reports an ablation over the temporal window size w .

Table 2: Effect of temporal window size w on QRC train MSE.

Window	Train MSE	Feature Dim
$w = 1$ (no window)	48.0	32
$w = 3$	32.6	96
$w = 5$	22.1	160

The 54% improvement from $w = 1$ to $w = 5$ is a direct realisation of the Takens delay-embedding principle [16]: a sliding window in feature space provides the regression with sufficient trajectory history to disambiguate states that share a single-step feature signature.

4.4 Multi-system validation

To test whether the QRC pipeline generalises beyond Lorenz, we apply the same architecture (no per-system tuning) to two additional canonical chaotic systems: the Rössler attractor ($a = b = 0.2, c = 5.7$) and Lorenz-96 with $N = 8$ variables ($F = 8$). 10 seeds per system.

Table 3: QRC performance across three chaotic systems (10 seeds each), with the architecture fixed at 5q, 2L, $w = 5$, $\alpha = 1$.

System	Dim	Train MSE	Test MSE
Lorenz	3	19.6 ± 3.6	3.1 ± 0.6
Rössler	3	0.17 ± 0.05	1.8 ± 0.1
Lorenz-96 ($N = 8$)	8	9.0 ± 1.1	12.4 ± 0.6

The same architecture adapts to all three attractors with sub-second training. Lorenz-96 is the hardest case in absolute MSE because the 8-dim state exceeds the 5-qubit encoding capacity,

but the training cost is unchanged. Low across-seed variance (relative std $\leq 19\%$ in all systems) confirms that the random-reservoir initialisation is not a confounding factor.

5 Discussion

5.1 Architectural argument for fixed reservoirs at NISQ scale

The diagnosis established in Section 4.1 bears directly on architecture choice. Three distinct mechanisms by which QRC avoids the QPINN failure modes are:

No quantum gradient. Because the reservoir is fixed, no gradient through the quantum circuit is required. Both barren-plateau gradient vanishing and finite-shot gradient estimation noise are absent by construction.

Closed-form readout training. Ridge regression provides a globally optimal solution in closed form. The under-parameterised optimisation pathology exhibited by QPINN training does not arise.

Explicit temporal context. The temporal-windowing readout provides the regression with $w \cdot 2^q$ feature dimensions per prediction; for $w = 5, q = 5$ this is 160 features versus QPINN’s 45 trainable parameters. The expressive capacity available to the classical readout is $\sim 4\times$ that of the variational quantum circuit at this scale.

These observations together explain the 81% MSE reduction observed in Section 4.2 as the consequence of architectural choice rather than a property of any one numerical experiment.

5.2 Classical ESN baseline context

A natural follow-up question is whether QRC’s advantage over QPINN reflects quantum-specific benefit or is a property of the *reservoir-computing* paradigm in general.

We additionally trained a classical Echo State Network (ESN) baseline under identical conditions (window $w = 5$, same train/test split, ridge regression with $\alpha = 1$, 5 seeds, $N = 500$ neurons, spectral radius 0.9, leaking rate 0.3).

The result is: ESN at this scale achieves train MSE 0.10 ± 0.00 and test MSE 2.08 ± 0.03 , both lower than QRC (17.1 ± 3.7 and 3.2 ± 0.6). At 5 qubits the quantum reservoir’s 32-dimensional feature space is dwarfed by the ESN’s 500 neurons, so the ESN’s accuracy advantage is unsurprising.

We interpret this finding as follows. The dominant factor separating QRC from QPINN is not the quantum-vs-classical character of the features but the *fixed-reservoir vs end-to-end variational* architecture choice. Both QRC and ESN benefit from closed-form ridge regression and temporal windowing; both vastly outperform the variational QPINN. The quantum reservoir does not provide an accuracy advantage at the 5-qubit scale — and theoretical considerations alone predict that it should not, since $2^5 = 32 \ll 500$.

5.3 Empirical scaling and the right scaling axis

The above suggests an experimentally tractable question: at what qubit count does the quantum reservoir’s exponential feature dimension overtake the classical ESN baseline at $N_{\text{ESN}} = 500$?

The naive feature-dimension-parity heuristic predicts the crossover at $q > \log_2 500 \approx 8.97$, i.e. $q = 10$ qubits give 1024 features that exceed the standard ESN baseline. The empirical measurement contradicts this prediction. We ran QRC at $q \in \{5, 6, 7, 8, 9, 10\}$ on the same Lorenz trajectory under matched training protocol, against the ESN baseline held fixed at $N_{\text{ESN}} = 500$. The result (Table 4) is that QRC test MSE *degrades monotonically* with q rather than improving:

The mechanism is signal-dilution. At fixed 3-dim target, the encoding pads the input vector to n qubits with zeros, so most basis-state amplitudes vary along a 3-dim subspace of the 2^q feature space. As q grows, the regression sees more features but not more signal — equivalently,

Table 4: Empirical scaling of QRC test MSE on Lorenz at fixed 3-dim target, 5 matched seeds per cell; ESN baseline at $N_{\text{ESN}} = 500$. QRC never overtakes the ESN baseline.

q	feature dim 2^q	QRC test MSE	ESN test MSE	winner
5	32	3.02 ± 0.63	0.38 ± 0.02	ESN ($p = 0.008$)
6	64	4.54 ± 0.59	0.38 ± 0.02	ESN ($p = 0.008$)
7	128	9.40 ± 4.85	0.38 ± 0.02	ESN ($p = 0.008$)
8	256	20.7 ± 11.4	0.38 ± 0.02	ESN ($p = 0.008$)
9	512	40.2 ± 18.3	0.38 ± 0.02	ESN ($p = 0.008$)
10	1024	68.7 ± 28.9	0.38 ± 0.02	ESN ($p = 0.008$)

the effective rank of the feature matrix saturates while its column count grows. Ridge regression with 50 training points and up to 5120 windowed features overfits the extra columns to training noise.

The implication is that “crossover at fixed target dimension” is the wrong scaling experiment. The correct experiment is *matched-complexity* ($K = q$): scale the prediction-target dimension together with the qubit count, so that each additional feature has a corresponding component of signal to explain. This matched-complexity protocol on Lorenz-96 shows QRC’s mean test MSE remaining flat across $q = N \in \{5, \dots, 11\}$ while ESN at matched feature dimension 2^q oscillates — the inverse of the trend reported above. We treat this empirically established distinction (signal-dilution at fixed target vs. matched-complexity scaling) as a methodological contribution in its own right, and defer the matched-complexity scaling result to the companion manuscript in preparation.

5.4 Information-symmetric ablation: WVQC

QPINN learns a global function $f_\theta(t) \rightarrow (x, y, z)$ from time alone; QRC learns a one-step transition map conditioned on a w -window of ground-truth prior states. This is an information asymmetry: QRC has access to recent trajectory history that QPINN does not.

The controlled ablation that removes this asymmetry is the *Windowed Variational Quantum Circuit* (WVQC): a variational quantum circuit with the same $w = 5$ windowed input as QRC, trained by Adam on a mean-squared-error loss against the next state. The encoding stage is a w -step data re-uploading sequence that applies the same per-component angle encoding as QRC’s input stage; the variational stage is identical to QPINN’s (4 qubits, 3 layers, 45 trainable parameters); the training protocol matches QPINN’s (Adam, 200 iterations, learning rate 10^{-3} with decay 0.995/step, adaptive gradient clipping at the 99-th percentile).

Table 5: Information-symmetric ablation. WVQC has the same windowed input as QRC and the same variational training as QPINN; 5 matched seeds.

Method	Train MSE	Test MSE
QPINN (4q, 3L, scalar-time input)	91.3 ± 21.9	47.9 ± 36.6
WVQC (4q, 3L, $w = 5$ windowed input)	61.7 ± 12.4	39.8 ± 14.6
QRC (5q, 2L, $w = 5$ windowed input + ridge)	17.1 ± 3.7	3.2 ± 0.6

The result disentangles the two effects:

- Giving the variational circuit the same windowed input as QRC reduces train MSE by approximately 32% (from 91.3 to 61.7). Windowed input does provide measurable help.
- However, WVQC remains $3.6\times$ above QRC on train MSE and $12.4\times$ above QRC on test

MSE, and its per-seed variance (± 12.4) is characteristic of the capacity-limited optimisation dynamics established in Section 4.1. The information asymmetry is not the dominant factor.

We conclude that the QRC–QPINN gap on this task is driven principally by the architectural choice between a fixed reservoir with closed-form ridge readout and an end-to-end variationally trained circuit, with input-information access making a secondary but real contribution. The capacity-vs-barren-plateau diagnosis of Section 4.1 extends to the windowed setting: providing the variational circuit with richer input does not rescue it from the capacity-limit failure mode, because the bottleneck is in the trainable parameter count, not the input.

5.5 Hardware deployment outlook

All experiments use exact statevector simulation (Qiskit Aer); no hardware runs were performed. The structural argument for QRC’s hardware-readiness is:

- **Fixed reservoir** requires no quantum gradient computation. Full training reduces to a noise-tolerant classical linear regression on readout features, parallelisable and hardware-agnostic.
- The **transverse-field Ising Hamiltonian** corresponds to a sparse ZZ -interaction graph plus single-qubit X rotations — a gate set native to IBM superconducting and IonQ trapped-ion platforms, minimising transpilation overhead.

QPINN on hardware would require gradient estimation via the parameter-shift rule [15] across thousands of iterations, with shot noise compounding training instability. Even in noiseless simulation, QPINN required several hours and did not converge cleanly.

Validating these projections on actual devices — and characterising the effect of depolarising noise on the QRC readout — is the priority future direction.

5.6 Limitations

- **Simulator-based.** All experiments use exact statevector simulation; hardware noise effects are deferred to a future revision.
- **Circuit scale.** We test small circuits (4–5 qubits); scaling behaviour at larger qubit counts is the subject of the scaling-crossover argument (Section 5.3) but is not yet experimentally validated.
- **Information asymmetry.** The QPINN–QRC comparison gives QRC access to a windowed history that QPINN lacks; the WVQC ablation (Section 5.4) addresses this directly.
- **QPINN budget.** The QPINN model did not converge within the 200-iteration budget. A larger budget would improve QPINN’s absolute performance but is not expected to change the qualitative diagnosis: 45 parameters cannot represent a chaotic trajectory regardless of gradient steps taken.

6 Conclusion

We have diagnosed the variational quantum failure mode on Lorenz chaotic prediction at NISQ scale as capacity-limited, not barren-plateau-driven. The McClean bound and direct gradient-norm measurements rule out exponential gradient suppression as the operative pathology at 4 qubits and 3 layers. The implication for the field is operational: at NISQ scale, mitigations targeted at gradient vanishing do not address the dominant failure mode, and architectural choices that bypass capacity limits (such as fixed-reservoir QRC) are the appropriate response.

QRC at matched quantum resources outperforms QPINN by 81% on train MSE, 93% on test MSE, and is $\sim 52,000\times$ faster to train. The same architecture generalises to Rössler and Lorenz–96 without hyperparameter retuning. We have additionally provided a heuristic prediction — crossover qubit count ≈ 10 — for when the quantum reservoir’s feature dimension would overtake the classical ESN baseline that currently wins at 5 qubits. This is the concrete experimental target to which the paper’s diagnostic argument naturally points.

Data and Code Availability

All code, training scripts, and analysis notebooks are available at https://github.com/pandey-tushar/Quantum_Chaos_solver.

References

- [1] E. N. Lorenz. Deterministic nonperiodic flow. *Journal of the Atmospheric Sciences*, 20(2):130–141, 1963.
- [2] M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks. *Journal of Computational Physics*, 378:686–707, 2019.
- [3] A. Pattnaik et al. Quantum-classical physics-informed neural networks for solving nonlinear differential equations. *arXiv:2503.16678*, 2024.
- [4] Y.-C. Leung et al. Quantum physics-informed neural networks for solving differential equations. *arXiv:2212.00360*, 2022.
- [5] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven. Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9(1):4812, 2018.
- [6] M. Cerezo et al. Variational quantum algorithms. *Nature Reviews Physics*, 3(9):625–644, 2021.
- [7] K. Fujii and K. Nakajima. Harnessing disordered-ensemble quantum dynamics for machine learning. *Physical Review Applied*, 8(2):024030, 2017.
- [8] P. Mújal et al. Opportunities in quantum reservoir computing and extreme learning machines. *Advanced Quantum Technologies*, 4(8):2100027, 2021.
- [9] K. Nakajima, K. Fujii, M. Negoro, K. Mitarai, and M. Kitagawa. Boosting computational power through spatial multiplexing in QRC. *Physical Review Applied*, 11:034021, 2019.
- [10] M. Kornjaca et al. Large-scale quantum reservoir computing with an analog quantum computer. *arXiv:2407.02553*, 2024.
- [11] S. Ahmed, L. Tennie, and L. Magri. Robust quantum reservoir computing for chaotic dynamics. *Proceedings of the Royal Society A*, 481:20250550, 2025.
- [12] X. Li et al. Quantum reservoir computing for realized volatility forecasting. *arXiv:2505.13933*, 2025.
- [13] H. Jaeger and H. Haas. Harnessing nonlinearity: Predicting chaotic systems and saving energy in wireless communication. *Science*, 304(5667):78–80, 2004.
- [14] J. Pathak, B. Hunt, M. Girvan, Z. Lu, and E. Ott. Model-free prediction of large spatiotemporally chaotic systems from data. *Physical Review Letters*, 120:024102, 2018.

- [15] K. Mitarai, M. Negoro, M. Kitagawa, and K. Fujii. Quantum circuit learning. *Physical Review A*, 98:032309, 2018.
- [16] F. Takens. Detecting strange attractors in turbulence. In *Dynamical Systems and Turbulence*, vol. 898 of *Lecture Notes in Mathematics*, pages 366–381. Springer, Berlin, 1981.